Online appendix: case studies and a neural procedure for proposing settings

to "Governing privacy risk across repeated statistical releases"

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This online appendix goes with the author manuscript "Governing privacy risk across repeated statistical releases: probability models, programme tolerances and accountable decisions". It holds material that the main text refers to but does not need: case studies from other data products, and a neural procedure for proposing candidate settings, with its variant SENTRY-R. Nothing here enters the results of the main text. Where this appendix refers to the main text it names the section or result rather than giving a number. Every numerical table here is produced by the packaged pipeline in the companion repository.

1 Case studies from other data products

The cases below are not pooled into one empirical estimate. They concern different data products, adversaries and harm classes, and the published evidence does not support a common denominator. Each case identifies which component of the proposed decision record would have made the public claim more precise. Table 1 records that mapping. Each row reads from left to right: what was released and how it was attacked, the object of the framework that describes the attack, and what the case teaches about the decision record. The last column carries the point of the table: in each case the public claim lacked a part of the record, such as a declared adversary, a programme guarantee, a decision owner or a stated residual risk. The first row is the credit register of the main text (the section on a granular credit register), treated there with measurements; the other four are not central bank cases and are described here without new measurements.

1.1 Sparse behavioural data: the Netflix Prize

The Netflix Prize data replaced direct identifiers with pseudonymous user numbers, but the released rating vectors were high-dimensional and sparse. Narayanan and Shmatikov [2008]

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Table 1: Case studies mapped to the framework

Case	Release and attack	Mathematical object	Governance lesson
Granular credit register	Loan-level file, aggregates and monthly revisions	Identification, predictive value of an alleged relationship, attribute and residual disclosure, differencing	One lever does not govern five risks; each needs its own target set and tolerance.
Netflix Prize	Sparse ratings linked to outside ratings	Posterior linkage p_{tj} and pooled auxiliary information	"Anonymous" identifiers do not define an adversary or bound posterior success.
US Census 2020	Published tabulations and database reconstruction	Programme composition and formal privacy accounting	Repeated exact statistics require a programme guarantee, not isolated marginal checks.
England and Wales Census 2021	Tables protected by record swapping and cell-key perturbation	Mechanism risk–utility frontier	Operational protection must be paired with an explicit scenario, uncertainty statement and decision owner.
HIPAA de-identification	Safe Harbor or Expert Determination	Rule-based versus scenario-based threshold	A legal route is not itself a numerical probability; Expert Determination makes assumptions and residual risk reviewable.

showed that a small amount of auxiliary rating information could link records across datasets. In the notation of the main text (the unified linkage experiment), the released identifier was irrelevant to the attack: Z_t contained a few ratings and approximate dates, L_{tj} measured their agreement with candidate records, and the largest posterior in the maximum-posterior rule of the main text determined the match. The case therefore illustrates the failure of exchangeability, not merely the presence of sample uniques.

The defensible retrospective test is not whether the file was anonymised. It is: for a prespecified distribution of auxiliary ratings and date error, what is the distribution of $\max_j p_{tj}$ across targets, and what simultaneous upper bound lies below the owned tolerance? A modern replication should vary the number of auxiliary ratings, their popularity, the date tolerance and the abstention threshold. Those variables define a family of attacks; choosing only an easy one after seeing the outcome would not validate the release.

1.2 Repeated official statistics: the 2020 US Census

The US Census Bureau adopted a disclosure avoidance system built on differential privacy [Abowd, 2018] after demonstrating that many published tabulations can support database reconstruction [Garfinkel et al., 2019, Abowd et al., 2022, 2023]. This is the institutional case closest to the proposition of the main text that marginal risks do not bound pooled risk. The threat does not arise because one table has a large marginal linkage risk. It arises because many accurate, overlapping constraints describe the same underlying population.

The framework makes two distinctions. First, reconstruction, membership inference and correct named-person linkage are different events and must not be reported under one risk label. Second, a formal privacy accountant governs the programme mechanism, whereas $\sum_i R_i$ governs only a union of declared single-release attacks. The policy debate over accuracy remains a risk–utility choice, but the protection claim and its unit of composition are now explicit. The appropriate empirical study would compare redistricting, small-area and demographic estimands under alternative privacy-loss budgets, then report subgroup-specific utility losses beside the programme guarantee. Ruggles et al. [2019] set out the costs of the change for social and economic research, and Kenny et al. [2021] find that the new system biases the population of mixed precincts used in redistricting.

1.3 Layered operational protection: Census 2021 in England and Wales

The Office for National Statistics describes protection of Census 2021 outputs through record swapping and the cell-key method, together with checks against disclosure in tables [ONS, 2022]. This is not evidence that the release has a universal linkage probability, nor need it be. It is evidence of a layered mechanism M_θ whose tuning changes both tabular accuracy and the information available to an intruder.

For governance, what is missing is a declared attack experiment. An evaluation could draw targets from separately stated population groups, expose the attacker to realistic public registers, reproduce the cell-key and swapping randomness, and calculate $\overline{R}_{1-\alpha}$ for each protected group. The mechanism parameters, the seeds policy, the table family and the expiry date would then be part of the governance threshold G defined in the main text. That turns a process description into a reproducible release decision, without implying that the mechanism offers differential privacy unless it does.

1.4 Health data: HIPAA Safe Harbor and Expert Determination

US health-data guidance recognises two routes for de-identification: removal of specified identifiers under Safe Harbor, and Expert Determination that the risk of identification is very small, using generally accepted statistical and scientific principles [HHS, 2012]. The contrast maps directly to this paper. Safe Harbor constrains the released variables; Expert Determination is a scenario-conditioned judgement about residual risk. Neither phrase alone fixes T, a, h, τ or α.

A complete determination would therefore file the population and target set, the accessible auxiliary sources, the linkage rule, the harm class, the estimate, the uncertainty bound and the review horizon. Where data are released repeatedly to the same recipient, the determination must count the accumulated products as part of what the adversary holds. This does not replace legal judgement. It makes the empirical premises of that judgement capable of expiry, challenge and replication.

1.5 A common case-study protocol

Across the five settings, a comparable prospective study has seven steps:

1. freeze the release mechanism, and define neighbouring or alternative confidential states where a formal guarantee is claimed;
2. specify the target population, the auxiliary information and the abstention rule;
3. separate linkage, membership, attribute and reconstruction success events;
4. simulate or execute attacks with the mechanism randomness and the shared data dependencies preserved;
5. estimate target and subgroup risks with simultaneous uncertainty bounds;
6. measure utility on prespecified authorised analyses, including subgroup error rather than only an aggregate score; and
7. assess the complete release programme, then file the governed tuple G.

The resulting studies would be comparable in design while keeping different numerical scales where the estimands genuinely differ.

2 An audited margin vector with a neural proposal model

A release decision needs a measured quantity, not a score. For a proposed mechanism θ, let $j = (a, h, g, v)$ index an adversary, harm class, target group and release vintage, including a separate index for an attacker pooling vintages. The finite, declared index set is $\mathcal{J}$, of cardinality n. A scenario j fixes the target sampling frame, attack algorithm, access to auxiliary information, release programme and binary definition of success. Conditional on a fixed proposed release and these specifications, let $P_{\theta j}$ be the law of one fresh attack experiment and define the actual risk $r_j(\theta) = \mathbb{E}_{P_{\theta j}} S_j$, where $S_j \in \{0, 1\}$ records success. The tolerance $\tau_j \in (0, 1)$ is fixed before the final audit. A population average and a particular target require distinct scenario definitions. A neural score does not supply this quantity. We call the vector of independently audited margins the *audited margin vector*, and its smallest entry the *minimum audited margin*:

$$\mathbf{z}(\theta) = \left(1 - \frac{\overline{r}_j(\theta)}{\tau_j}\right)_{j \in \mathcal{J}} \in \mathbb{R}^n, \qquad T(\theta) = \min_{j \in \mathcal{J}} z_j(\theta). \tag{1}$$

Here $\overline{r}_j$ is a simultaneous upper confidence bound, with its sampling unit and adversary written into the assessment. A nonnegative T means every declared coordinate passes. The complete vector must accompany the scalar: the minimum alone conceals which group or programme attack constrains the release. Utility $U(\theta)$ is kept as a separate coordinate, since a release can have large audited margins while carrying little public value. Missing or unaudited coordinates receive $\overline{r}_j = 1$ and therefore fail whenever $\tau_j < 1$. Each assessment thus starts with no inherited claim of safety: collect independent evidence, tighten the bounds, improve the mechanism, and recertify. It does not promise that evidence will rescue an unsafe release.

Figure 1 depicts the coordinatewise decision. Each entry is a separately audited margin, and one negative entry forces the minimum below zero even if the other coordinates pass.

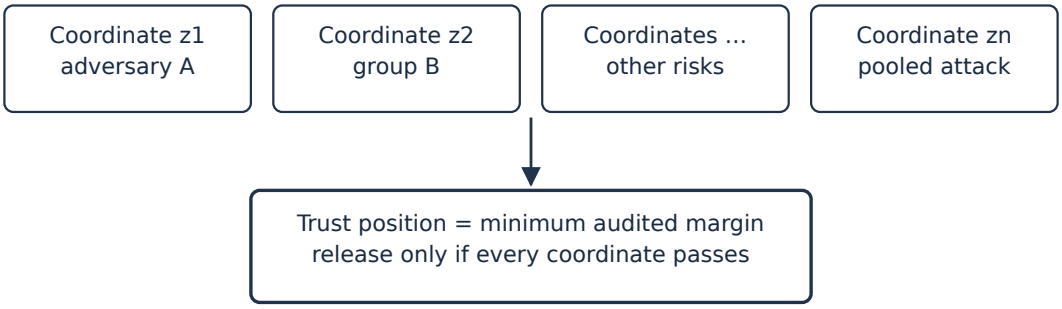


Figure 1: The n-dimensional trust position uses independently bounded risk coordinates $z_j = 1 - \overline{r}_j/\tau_j$. The labels illustrate declared coordinates, not numerical results.

2.1 The neural model and what its output means

Encode a proposed release and its environment as $x(\theta) \in \mathbb{R}^d$: mechanism parameters, sample fraction, key entropy and sparsity, auxiliary information quality, number and overlap of prior releases, access controls, population shifts and target-group indicators. All features must be available *before* certification and their provenance recorded. A transparent baseline is a regularised logistic model; our nonlinear proposal model is

$$H_0 = x, \qquad H_{\ell+1} = \rho(H_\ell W_\ell + b_\ell), \quad 0 \le \ell < L, \tag{2}$$

$$\widehat{\boldsymbol{r}}_\phi(x) = \sigma(H_L W_{\text{out}} + b_{\text{out}}) \in (0,1)^n, \quad \mathcal{L}(\phi) = -\sum_{i,j} w_{ij}[y_{ij}\log\widehat{r}_{ij} + (1-y_{ij})\log(1-\widehat{r}_{ij})]. \tag{3}$$

The `neural_trust` module fixes $L = 1$; `sentry_r` makes L configurable. The label y_{ij} is the *observed success of an attack* on a separate scenario draw, with a zero label for a failed or abstaining attack as specified in the threat model. Unevaluated pairs have $w_{ij} = 0$; repeated attacks sharing one auxiliary file are clustered by file rather than treated as independent. The sigmoid outputs are predicted attack frequencies, not audited margins or probabilities that a release is safe, and no ordering of them across harm classes is implied. Train on past mechanism–adversary scenarios, hold out whole populations, release programmes and vintages, and keep the final audit draws out of all training, tuning, calibration and feature selection. Compare log loss, Brier score and reliability diagrams with the logistic and frequency baselines, both overall and by group [Guo et al., 2017]. A calibration plot is a diagnostic, never a certificate for the highest-risk individual.

The neural model proposes a *finite* shortlist Θ_0: it ranks candidates by their largest predicted risk-to-tolerance ratio; utility enters only the final choice among certified candidates. For each candidate and j, run $B_{\theta j}$ fresh, conditionally identically distributed Bernoulli attacks and count $K_{\theta j}$ successes. Fix the sample sizes before observing final outcomes. With $M = |\Theta_0|n$ prespecified inspected coordinates, define the exact one-sided bound

$$\overline{r}_{\theta j} = \begin{cases} 1, & K_{\theta j} = B_{\theta j}, \\ F^{-1}_{\text{Beta}(K_{\theta j}+1,\, B_{\theta j}-K_{\theta j})}(1-\alpha/M), & K_{\theta j} < B_{\theta j}. \end{cases} \tag{4}$$

Certification uses (1) and (4), never the neural output. The final choice may maximise U among certified candidates, provided the shortlist was fixed without looking at these final audit draws.

Figure 2 makes the separation between development and final evidence explicit. The arrows into the final gate carry audit bounds and tolerances; the network contributes only the fixed shortlist.

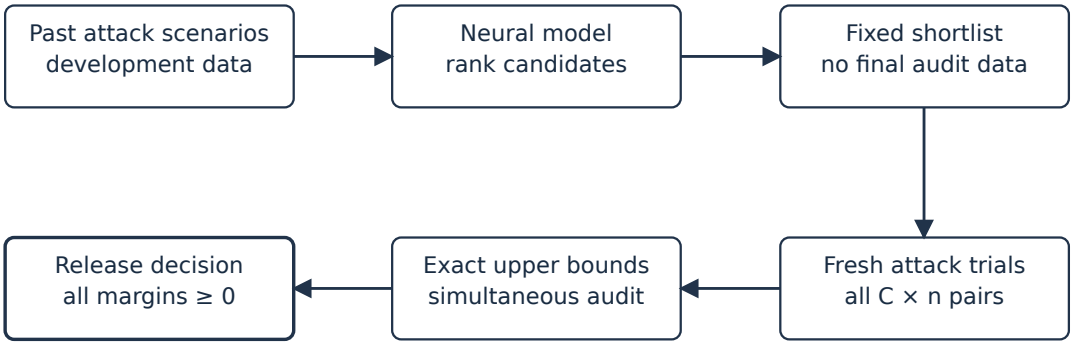


Figure 2: The neural network ranks candidates; independent attack trials and exact simultaneous upper bounds decide. $C = |\Theta_0|$ and n is the number of risk coordinates.

Proposition 1 (Selection-safe finite-family certification)**.** *Condition on development information $\mathcal{D}$ and its resulting finite shortlist, candidate releases, scenarios and sample sizes. Suppose each final audit count is* $\mathrm{Binom}(B_{\theta j}, r_j(\theta))$ *conditional on $\mathcal{D}$; auxiliary information sampled afresh is included in this experiment, whereas fixed auxiliary information is conditioned upon. Then with probability at least $1-\alpha$, simultaneously for every inspected pair, $r_j(\theta) \le \overline{r}_{\theta j}$. Consequently the rule selecting any candidate with $T(\theta) \ge 0$ has probability at most α of certifying a candidate whose risk exceeds a declared tolerance in any coordinate.*

Proof. For one pair the event that the exact upper bound falls below the true risk has probability at most α/M by inversion of the binomial lower-tail test (Appendix 4). A union bound over the M pairs gives the simultaneous statement, without requiring independence between pairs. On this event every certified candidate satisfies $r_j(\theta) \le \tau_j$ for all j, regardless of which certified candidate the final rule selects. Taking expectation over $\mathcal{D}$ gives the unconditional guarantee. □

If the release, attack, shortlist or group definitions change in response to final outcomes, the conditional argument no longer applies. Freeze and restart with untouched audit draws, or replace the fixed-sample bound by the anytime procedure of the main text (the section on adaptive attacks and anytime-valid certification) with an explicit multiplicity allocation. Binomial sampling covers attack-draw variability conditional on a fixed population and mechanism. It does not cover uncertain input data, unknown auxiliary sources, a stronger future attacker or distribution shift. Those are separate scenario coordinates or require a stated outer uncertainty analysis.

2.2 What changes as the dimension grows

The number d of input features and the number n of risk coordinates are different. For widths $q_0 = d, q_1, \ldots, q_L, q_{L+1} = n$, a dense forward pass costs $O(\sum_{\ell=0}^{L} q_\ell q_{\ell+1})$ operations per scenario and stores $O(\sum q_\ell q_{\ell+1})$ parameters; a training epoch multiplies the cost by the number of labelled scenarios. Sparse embeddings or shared encoders can reduce cost when many groups share structure, but cannot supply evidence in an unseen group. A full final audit over C candidates and n coordinates uses Cn confidence bounds and, at zero observed successes in every coordinate, needs at least

$$B \geq \left\lceil \frac{\log(\alpha/(Cn))}{\log(1-\tau_{\min})} \right\rceil \tag{5}$$

independent trials per coordinate to certify a common minimum tolerance $\tau_{\min}$. This is necessary and sufficient when that coordinate has zero successes; with successes, more trials can be needed. Thus computational amortisation by a network does not erase the statistical cost of many simultaneous guarantees. For an L-Lipschitz risk function on $[0,1]^d$, even a direct grid at accuracy ϵ uses order $(L/\epsilon)^d$ cells. Without such regularity, finite observations alone cannot guarantee uniform pointwise accuracy. Report learning curves against both d and n; do not describe merely adding hidden units as solving this problem.

For stress testing, if the full law $Q_{\theta j}$ of the deployed attack experiment is known to satisfy $\mathrm{TV}(P_{\theta j}, Q_{\theta j}) \leq \delta_j$, bounded attack success gives $r_{Q_{\theta j}}(\theta) \leq r_{P_{\theta j}}(\theta) + \delta_j$. A declared and justified bound can therefore replace $\overline{r}_j$ by $\min(1, \overline{r}_j + \delta_j)$. Estimating δ_j from a neural embedding alone does not justify that guarantee. Out-of-scope feature values, new linkage data and an unbounded shift require refusal and a new assessment.

The testable methodological contribution is the combination of an audited margin vector, a neural search model, and an independent simultaneous audit under adaptive choice of the final candidate. Neural prediction, binomial bounds and neural calibration individually have extensive precedents [Bates et al., 2021, Guo et al., 2017]; the claim here is about their stated division of labour for a release programme, not an untested claim of superior empirical performance. The accompanying `privacy_risk.neural_trust` module includes a seeded synthetic demonstration of the search and audit interface. It cannot validate the existing tables in this paper. An empirical extension must preregister targets and populations, compare the network with the logistic and frequency baselines at fixed audit budgets, report selective utility and worst-group failures, and rerun every pooled attack on held-out real programmes.

3 SENTRY-R: scenario-enriched neural triage for release review

We name the implementable extension *SENTRY-R*, shorthand for scenario-enriched neural triage for release review. This names the specified workflow, not a claim that its components are individually new or that a fitted network has improved real release decisions. Its separation of three data streams is central: *untrusted scenario proposals*, *measured development attacks*, and *untouched final audits*. The implementation supports any finite input dimension d and any finite number n of declared risk coordinates. A deeper network predicts attack frequencies for

search; it does not certify a release.

3.1 LLM-assisted scenario expansion and evidence provenance

Let $\mathcal{F}_t$ denote the information available before development round t, including past mechanisms, validated scenario specifications and measured attacks. A language model proposes descriptions and numeric scenario features $Q_t = G_t(\mathcal{F}_t, \xi_t)$. A deterministic validator checks the predeclared feature names and bounds and the scenario identifier. The calling code, not the language model, writes the source metadata: the model name and the SHA-256 hash of the exact prompt. The validator does not treat free-form text or a language-model risk estimate as an observed success. A separate attack evaluator then draws Z_{ti} under the accepted scenario and records $K_{tj} = \sum_{i=1}^{B_{tj}} s_j(Z_{ti})$, together with its experiment identifier and sampling unit. The retraining history grows by

$$\mathcal{F}_{t+1} = \mathcal{F}_t \vee \sigma\big(Q_t, \mathrm{valid}(Q_t), \{K_{tj}, B_{tj}, \mathrm{provenance}_{tj}\}_{j\in\mathcal{J}}\big)\,. \tag{6}$$

The language model can change which *development* experiment is run next; it does not directly change a network weight or label. Source identifiers, scenario hashes and an explicit partition label make the update inspectable. Treating repeated targets or auxiliary files as independent Bernoulli trials would break the specified sampling model. Human review is still required to establish whether a proposed adversary, harm definition and auxiliary source are plausible and permissible.

Let $\mathcal{D} = \mathcal{F}_{t_*}$ be all such development information at a chosen stopping time t_*, including model weights and hyperparameter selection. A separate validation set holds out entire release programmes; final audit draws are excluded from $\mathcal{D}$. Conditional on $\mathcal{D}$, freeze the coordinate definitions, policy tolerances, C shortlisted candidates, the family error α and the planned sample sizes $B_{\theta j}$. The implementation writes all of these, with a named tolerance owner, an expiry date and every development evidence identifier, into one hashed freeze record. The final audit reads α from that record and refuses to run if the record has changed, if the expiry date has passed, if any trial count differs from the plan, or if an audit evidence identifier was used in development. An optional append-only ledger refuses a second audit of the same freeze. These checks bind an audit to its freeze; they do not prove the order of events, since the hash has no key or trusted time stamp and a new freeze or a new ledger can always be written. The Proposition 1 guarantee then applies even if the *development* sequence used an LLM adaptively, provided the conditional binomial model for the untouched final draws holds. A final audit reused as development data, or an LLM proposal generated from its outcomes, invalidates that conditioning. Figure 3 shows the implemented boundary. It reads from left to right and then down. A scenario proposed by a language model passes a fixed schema and a named review, and it can add a scenario but never an outcome. Only counted development attacks train the deep proposal model, whose ranking fixes a shortlist of candidate settings. Only the fresh final audit, with simultaneous bounds over every coordinate, reaches the release decision. The note under the first box states the rule the figure is there to show: proposal text never supplies a training label or a release certificate.



Figure 3: SENTRY-R development and decision boundary. The LLM's scenario proposal is checked against a fixed schema; only measured development attacks enter the deep model. Its ranking freezes a shortlist before a fresh, independent final audit.

3.2 A deep model trained on counted attack outcomes

The `privacy_risk.sentry_r` module provides a depth-L fully connected ReLU model with common width q, including the depth-zero logistic baseline. Training features $x_i \in \mathbb{R}^d$ are standardised using *training* means and scales only. With $h_i^0 = x_i$, for $\ell = 0, \ldots, L-1$ define

$$h_i^{\ell+1} = \rho(h_i^\ell W_\ell + b_\ell), \quad a_{ij} = \left(h_i^L W_\mathrm{o} + b_\mathrm{o}\right)_j, \quad p_{ij} = \sigma(a_{ij}), \quad \rho(z) = \max\{0, z\}. \tag{7}$$

For each observed development pair (i, j) let $K_{ij} \in \{0, \ldots, B_{ij}\}$ be the count of successes in $B_{ij} \geq 1$ trials. The implementation minimises the binomial negative log likelihood, without its constant combinatorial term, and a ridge term:

$$\mathcal{L}(\phi) = -\frac{1}{B_\bullet} \sum_{i,j} \left[K_{ij} \log p_{ij} + (B_{ij} - K_{ij}) \log(1 - p_{ij})\right] + \frac{\lambda}{2} \sum_{\ell=0}^{L} \|W_\ell\|_F^2, \quad B_\bullet = \sum_{i,j} B_{ij}. \tag{8}$$

Here $W_L = W_\mathrm{o}$; a weight attached to trial count is appropriate only if the count's sampling design was fixed without peeking at those outcomes. To see exactly what is learned, assume $K \mid x, B \sim \mathrm{Binom}(B, \pi_j(x))$ with B determined before outcomes. For a fixed x, B, the expected unregularised loss minus its value at $p = \pi_j(x)$ is

$$\frac{B}{B_\bullet} D_{\mathrm{KL}}\left(\mathrm{Bern}(\pi_j(x)) \,\|\, \mathrm{Bern}(p)\right) \geq 0. \tag{9}$$

Thus the ideal unrestricted conditional predictor is the attack success frequency $\pi_j(x)$ on the *development* distribution; finite data, regularisation, optimiser error and population shift prevent equality in an actual trained model. No neural output is a confidence limit on an individual or a new population.

The model is fitted by backpropagation. With $P = (p_{ij})$, $K = (K_{ij})$ and $B = (B_{ij})$, the output-logit derivative is $(B \odot P - K)/B_\bullet$, and the hidden-layer gradients follow by the chain rule in the usual way [Goodfellow et al., 2016, ch. 6]. The package includes a finite-difference check of the derivatives.

3.3 Training, scale and falsification

The network uses seeded mini-batches, Adam, gradient clipping and early stopping on a separate held-out programme. Depth zero is trained on the same counted outcomes as a logistic baseline. For N development rows and E epochs, the dense leading arithmetic order is $O\{EN[dq + (L - 1)q^2 + qn]\}$ for $L \geq 1$, with $O\{dq + (L - 1)q^2 + qn\}$ weights; the logistic case has $O(ENdn)$ arithmetic. This is a computational scaling statement, not a statistical sample-size promise. The final Cn confidence bounds and independent attack trials remain necessary whatever the prediction speed. Unknown groups and out-of-domain features trigger a new assessment rather than an extrapolated certificate.

A valid empirical evaluation must compare deep and logistic rankings at fixed shortlist sizes and audit budgets on held-out *real* programmes; report log loss, Brier score, worst-group calibration, selected utility and false gates; repeat pooled-vintage attacks; and show uncertainty across programmes. This package currently provides a seeded synthetic end-to-end demonstration, not such an evaluation. Table 2 is regenerated by `scripts/sentry_r_benchmark.py`. For each of ten seeds it draws three disjoint programmes from one toy law: 320 training rows, 100 rows used only for early stopping, and 2,000 test rows used only for the reported loss. Both models train until 150 epochs pass without improvement, and no best epoch reaches the cap. Trained to convergence, the logistic model has a slightly lower test log loss than the three-layer model in all ten replicates. The mean paired gap is smaller than the spread of either model across replicates. The table shows that the pipeline runs, not that depth helps. The test programmes come from the same law and feature range as the training data, so they do not test shift.

Table 2: Synthetic SENTRY-R comparison over 10 seeded replicates of the toy law. Each replicate draws 320 training rows, 100 rows used only for early stopping (patience 150 epochs, cap 6000) and 2000 test rows used only for the reported losses. Entries are mean (standard deviation) across replicates; smaller is better. The last row is the paired deep-minus-logistic test log loss (range 0.0004 to 0.0050; deep lower in 0 of 10). These are not results on real release programmes.

Proposal model	Test log loss	Test Brier score	Best epoch (range)
Logistic	0.2556 (0.0043)	0.0727 (0.0017)	404–898
Three-layer ReLU	0.2576 (0.0047)	0.0735 (0.0018)	79–402
Deep minus logistic	0.0020 (0.0014)		

4 Mathematical details for the audited margin vector

Exploratory real-data runs. The repository also holds a state-held-out pilot of the proposal model on the public Census file and a later comparison of linear, neural and quadratic proposers on the same file (`scripts/realdata_neural_pilot.py` and `scripts/realdata_neural_comparison.py`). Their linear baselines were not trained to convergence and their differences lie within bootstrap intervals, so they are not evidence for or against the procedure, and nothing in this paper rests on them.

Experiment and estimand. Let $\mathcal{D}$ contain all development observations, candidate generation, neural fitting, tuning, scenario definitions and proposed releases. Conditional on $\mathcal{D}$, the shortlist Θ_0, its cardinality C, and all sample sizes $B_{\theta j} \geq 1$ are fixed. An audit draw $Z_{\theta ji}$ includes the target, auxiliary data when freshly sampled, and any randomness in the declared attack. The measurable success indicator $S_{\theta ji} = s_{\theta j}(Z_{\theta ji})$ is one exactly when the declared harm criterion is met. For fixed (θ, j) assume the draws are identically distributed and independent conditional on $\mathcal{D}$, with law $P_{\theta j}$. Thus

$$r_j(\theta) = \int s_{\theta j}(z)\, dP_{\theta j}(z), \qquad K_{\theta j} = \sum_{i=1}^{B_{\theta j}} S_{\theta ji} \mid \mathcal{D} \sim \mathrm{Binom}(B_{\theta j}, r_j(\theta)). \tag{10}$$

The risk is conditional on a frozen release and on the audit design. If target records or auxiliary files are reused, the resulting dependent observations must be treated as one sampling unit or justified by another valid confidence procedure. Independence across distinct (θ, j) pairs is unnecessary. For a finite population sampled without replacement, the binomial assumption is inappropriate unless the attack design supplies independent replacement draws; a hypergeometric or design-based bound is needed instead.

Exact inversion. Write $b = B_{\theta j}$, $k = K_{\theta j}$, $\gamma = \alpha/(Cn)$, and $F_p(k) = \sum_{t=0}^{k} \binom{b}{t} p^t (1-p)^{b-t}$. For $k < b$, $F_p(k)$ is continuous and strictly decreasing in $p \in (0,1)$. The upper endpoint $u(k)$ is the unique solution of $F_{u(k)}(k) = \gamma$; set $u(b) = 1$. The binomial–beta identity gives

$$F_p(k) = 1 - I_p(k+1, b-k), \qquad u(k) = F^{-1}_{\mathrm{Beta}(k+1,b-k)}(1-\gamma). \tag{11}$$

For any fixed true p, the event $\{u(K) < p\}$ is a lower-tail set of integer counts: by monotonicity it equals $\{F_p(K) < \gamma\}$, apart from the harmless equality boundary. If k_* is the largest count in this set, $\Pr_p\{u(K) < p\} = F_p(k_*) < \gamma$; if the set is empty the probability is zero. Hence $\Pr_p\{p \leq u(K)\} \geq 1-\gamma$ for every $p \in [0,1]$. The possible strict inequality reflects the discreteness of the count.

Familywise selection. Define the event

$$E = \bigcap_{\theta \in \Theta_0} \bigcap_{j \in \mathcal{J}} \{r_j(\theta) \leq \overline{r}_{\theta j}\}. \tag{12}$$

Conditional on every development history for which the assumptions hold, $\Pr(E^c \mid \mathcal{D}) \leq \sum_{\theta,j} \gamma = \alpha$. Any measurable decision rule $\widehat{\theta}$ using the complete vector of final bounds, including the choice of the greatest utility among accepted candidates, satisfies the deterministic implication

$$E \implies \left[\widehat{\theta} \text{ accepted}\right] \implies \max_{j \in \mathcal{J}} \frac{r_j(\widehat{\theta})}{\tau_j} \leq 1. \tag{13}$$

The probability of a false certification is therefore at most α conditionally and, by $\Pr(E^c) = \mathbb{E}[\Pr(E^c \mid \mathcal{D})]$, unconditionally. This is a frequentist guarantee for the declared finite family; it is not a posterior probability that a particular accepted release is safe. An audit outcome

used to add candidates, alter scenarios, retrain the network or select sample sizes destroys the conditioning argument unless a separate sequential or adaptive design is proved valid.

Budget and an illustrative calculation. At $k = 0$, equation (11) simplifies to $u(0) = 1 - \gamma^{1/b}$. Consequently $u(0) \le \tau$ if and only if $(1-\tau)^b \le \gamma$, equivalently

$$b \ge \left\lceil \frac{\log \gamma}{\log(1-\tau)} \right\rceil. \tag{14}$$

For illustration only, $C = 4$, $n = 2$, $\alpha = 0.05$ and $\tau = 0.12$ give $\gamma = 0.00625$ and $b = 40$. The zero-success endpoint is approximately 0.11916 at $b = 40$, but approximately 0.12202 at $b = 39$. These numbers do not establish the risk of any real release. The total number of attack trials is $\sum_{\theta,j} B_{\theta j}$; when each of Cn pairs uses the same b, it is Cnb. For small τ, $-\log(1-\tau) = \tau + O(\tau^2)$, so the zero-success requirement grows as $\log(Cn/\alpha)/\tau$ per pair. The neural output dimension changes this multiplicity even when its input feature dimension d stays fixed.

Neural objective and its limits. Let $\pi_j(x) = \Pr(Y_j = 1 \mid X = x)$ in the development distribution. For a positive weight $w_j(x)$ depending only on x and j, the conditional expected binary cross entropy at a predicted probability q is

$$\ell_x(q) = -w_j(x)[\pi_j(x)\log q + (1-\pi_j(x))\log(1-q)], \quad \ell_x'(q) = w_j(x)\frac{q-\pi_j(x)}{q(1-q)}, \tag{15}$$

with $\ell_x''(q) > 0$ at its unique minimum $q = \pi_j(x)$ whenever $0 < \pi_j(x) < 1$. Outcome-dependent weights instead change that minimiser. The implemented one-hidden-layer ReLU model is fitted by ordinary backpropagation of this objective [Goodfellow et al., 2016, ch. 6]; the gradients are textbook ones, and fitting does not make the predictions exact. Write W_1 and W_2 for its fitted input-to-hidden and hidden-to-output weight matrices. Then ReLU's Lipschitz constant of one and sigmoid's scalar derivative at most $1/4$ give the predictor sensitivity bound $\|\widehat{\boldsymbol{r}}(x) - \widehat{\boldsymbol{r}}(x')\|_2 \le \|W_2\|_2\|W_1\|_2\|x - x'\|_2/4$. It controls changes in *predictions*, not errors relative to actual risks, and is no substitute for the audit.

The input dimension. Suppose, as an additional assumption, the true conditional risk is L-Lipschitz in $\|\cdot\|_\infty$ on $[0,1]^d$. A grid of spacing $h \le \epsilon/L$ lets a value at a nearby grid point approximate the true risk to within ϵ, requiring at most $(1 + \lceil L/\epsilon \rceil)^d$ grid points. Estimating those values adds sampling cost at each point; the grid count alone already grows exponentially in d. If no regularity links unsampled inputs to observed inputs, two risk functions can agree on every observed feature vector and disagree arbitrarily at an unobserved vector, so there is no distribution-free uniform interpolation guarantee. Structural assumptions may help prediction, but the finite-family audit guarantee above does not rely on them.

Distribution shift. Using $\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)|$ on the complete audit-draw space, the layer-cake representation for any success function $0 \le s \le 1$ yields $|\mathbb{E}_Q s - \mathbb{E}_P s| \le \mathrm{TV}(P,Q)$. If a justified deterministic bound $\mathrm{TV}(P_{\theta j}, Q_{\theta j}) \le \delta_{\theta j}$ holds for all inspected pairs, a simultaneous

deployed-risk bound is

$$\overline{r}_{\theta j}^{\text{deploy}} = \min\{1, \overline{r}_{\theta j} + \delta_{\theta j}\}, \qquad T^{\text{deploy}}(\theta) = \min_j \left(1 - \frac{\overline{r}_{\theta j}^{\text{deploy}}}{\tau_j}\right). \tag{16}$$

The same familywise probability applies because the TV inequality holds pointwise on the simultaneous coverage event. A TV estimate, itself uncertain, needs its own valid upper bound and error allocation. A distance between neural feature embeddings alone cannot establish a TV bound on the full experiment.

5 SENTRY-R derivations and adaptive development

Gradients. The depth-L network is fitted by ordinary backpropagation of the count loss (8). The only non-standard term is the output-logit derivative, $b\sigma(a) - k$ for a count of k successes in b trials, which follows from the chain rule; the rest is textbook [Goodfellow et al., 2016, ch. 6], and the package checks the derivatives by finite differences. Adam changes the update rule, not the gradients.

Adaptive proposal rounds and a fixed final experiment. A language model, human analyst or hyperparameter search may use the measured *development* history repeatedly. Let T be an almost surely finite stopping time for that history, and condition on $\mathcal{D} = \mathcal{F}_T$. Let the now fixed finite family have $M(\mathcal{D}) = C(\mathcal{D})n(\mathcal{D})$ positive coordinates, with $\gamma(\mathcal{D}) = \alpha/M(\mathcal{D})$. Assume for every fixed pair that its *new final* count has conditional law $K_{\theta j} \mid \mathcal{D} \sim \text{Binom}(B_{\theta j}, r_j(\theta))$, where $B_{\theta j}$, the release, tolerance, sampling frame and attack were fixed by $\mathcal{D}$ and the final draws were not used to construct it. By the exact inversion of Appendix 4,

$$\Pr\{r_j(\theta) > \overline{r}_{\theta j} \mid \mathcal{D}\} \le \gamma(\mathcal{D}), \qquad \Pr\{\exists(\theta, j) : r_j(\theta) > \overline{r}_{\theta j} \mid \mathcal{D}\} \le \alpha. \tag{17}$$

The second inequality is a union bound *conditional* on the complete adaptive development history; the number and identity of pairs may depend on that history. On its complementary event, every accepted candidate satisfies every declared tolerance. Integrating over $\mathcal{D}$ proves an unconditional error probability no greater than α. No claim follows for scenarios omitted from the declared family or for a final audit that has influenced development. In particular, feeding a final audit result back to an LLM and then trying another candidate with the same nominal α is an unaccounted additional selection step. The code refuses a second audit of one freeze when a ledger is kept, but it does not stop a new freeze after an audit has been seen.

Why measured trials cannot be replaced by language-model labels. The proper-scoring identity in (9) assumes $\mathbb{E}[K_{ij} \mid X_i = x, B_{ij} = b] = b\pi_j(x)$. Replacing K_{ij} by a model-generated estimate $\widetilde{K}_{ij}$ changes the minimiser to $\mathbb{E}[\widetilde{K}_{ij}/B_{ij} \mid X_i = x]$, which need not equal $\pi_j(x)$. A prompt that asserts a small risk is therefore no statistical evidence of small risk, even when its prose is plausible. The software refuses generated success counts, sets the proposal's model name and prompt hash itself, and refuses a final audit that reuses a development evidence

identifier. These checks rest on labels the caller supplies, so source integrity and experimental independence still require institutional checks.

Sensitivity versus validity. The network acts on standardised inputs $u_k = (x_k - \mu_k)/s_k$, with training means μ_k and scales $s_k > 0$, so $\|u - u'\|_2 \le \max_k s_k^{-1} \|x - x'\|_2$. For a fixed network with widths q_m, the spectral-norm bound in raw features is

$$\|\widehat{\boldsymbol{r}}(x) - \widehat{\boldsymbol{r}}(x')\|_2 \le \frac{1}{4} \left(\prod_{m=1}^{L+1} \|W_m\|_2 \right) \max_k s_k^{-1} \, \|x - x'\|_2. \tag{18}$$

It uses ReLU's unit Lipschitz constant and sigmoid's derivative at most $1/4$. The bound is for *predictions* at fixed weights. It provides no bound on $\|\widehat{\boldsymbol{r}} - \boldsymbol{r}\|$ or on risk under new auxiliary information. The certification statement is instead obtained from fresh final audit data and the declared experimental law.

6 Multiscale proposal models and audited distribution shift

This extension separates three mathematical objects: the unknown attack law, the fitted proposal function, and a confidence bound from new attack trials. Its results do not replace the registered sequential gate. In particular, a derivative of a neural network describes its response to an input change; it does not bound an adversary's true success probability. The new software is `privacy_risk.multiscale`. All new numerical demonstrations use synthetic laws and do not modify the earlier public-data results.

6.1 An arbitrary dimensional risk vector

Let $x \in \mathcal{X} = \prod_{j=1}^{d} [\ell_j, u_j]$ describe a release and its registered threat model, with $u_j > \ell_j$. For coordinate $i \in \{1, \dots, m\}$, let $A_i(Z, x) \in \{0, 1\}$ be the declared attack success event. Define

$$r_i(x; P_i) = \int A_i(z, x) \, dP_i(z), \qquad z_i(x) = 1 - r_i(x; P_i)/\tau_i, \quad 0 < \tau_i < 1. \tag{19}$$

A coordinate includes its target population, information available to the attacker, attack algorithm and harm. These components cannot change silently when the feature vector changes. Eligibility is the intersection of all registered inequalities $r_i \le \tau_i$, rather than an average over harms. The scalar $\min_i z_i$ records the limiting margin, but a report retains the whole vector so that a favourable component cannot conceal another violation.

6.2 Finite functional wavelet dictionary

Normalize $v_j = (x_j - \ell_j)/(u_j - \ell_j)$ and define

$$\psi(t) = (1 - t^2) e^{-t^2/2}, \qquad \psi'(t) = (t^3 - 3t) e^{-t^2/2}. \tag{20}$$

For levels $a = 0, \dots, J-1$ and centres $c_{ak} = (k+1/2)2^{-a}$, $k = 0, \dots, 2^a - 1$, the feature map is

$$\Phi_J(x) = \big(v_j, \psi(2^a(v_j - c_{ak}))\big)_{j,a,k} \in \mathbb{R}^D, \qquad D = d\,2^J. \tag{21}$$

The dictionary captures localized changes at several scales and retains the normalized identity coordinates. It is neither an orthogonal wavelet transform nor an exact interpolant. On a bounded interval the atoms need not be orthogonal. An exact interpolation claim would require specified nodes, an invertible collocation matrix and a conditioning estimate, none of which follows from using a wavelet-shaped activation. Tensor products of all atoms cost exponentially in d; the coordinatewise dictionary avoids that construction. Cross-feature interactions arise in the dense network. Its computational benefit is a choice of representation, not a universal approximation-error guarantee.

Put $s_j = (u_j - \ell_j)^{-1}$. Each feature depends on a single raw input, so the feature Jacobian has orthogonal columns. Since $|\psi'| \le C_\psi = 3\sqrt{3}e^{-3/2} + 3e^{-1/2}$,

$$\|D\Phi_J(x)\|_2 \le L_\Phi = \max_j s_j \left(1 + C_\psi^2 \sum_{a=0}^{J-1} 2^a 4^a\right)^{1/2}. \tag{22}$$

The bound follows by separately maximizing $|t|^3 e^{-t^2/2}$ and $3|t|e^{-t^2/2}$. Integration on a line segment in the convex box establishes $\|\Phi_J(x) - \Phi_J(y)\|_2 \le L_\Phi \|x - y\|_2$. Increasing resolution increases L_Φ and may worsen robustness; fine scales are not automatically an improvement.

6.3 Deep counted likelihood and explicit sensitivities

For column-vector notation let $h_0 = \Phi_J(x)$, $h_l = \mathrm{ReLU}(W_l h_{l-1} + b_l)$, $l = 1, \dots, L$, and $f_\theta(x) = \sigma(W_{L+1}h_L + b_{L+1}) \in (0,1)^m$. For development counts k_{ni} out of b_{ni} trials, fit

$$\mathcal{L}(\theta) = -\frac{1}{\sum_{n,i} b_{ni}} \sum_{n,i} \{k_{ni} \log f_{ni} + (b_{ni} - k_{ni}) \log(1 - f_{ni})\} + \frac{\lambda}{2} \sum_l \|W_l\|_F^2. \tag{23}$$

Binomial counts retain different trial denominators. Treating all observed fractions as equally precise generally changes this objective. If the counts are dependent, the expression is a working loss, not an independent binomial likelihood. Training, early-stopping validation and final audits must use separate groups; neither LLM proposals nor final certification trials provide training labels.

Away from ReLU kinks the raw-input derivative is

$$Df_\theta(x) = D_\sigma W_{L+1} D_L W_L \cdots D_1 W_1 D\Phi_J(x), \tag{24}$$

where D_l contains activation indicators and $D_\sigma = \mathrm{diag}(f_i(1 - f_i))$. The implementation stores transposed weights because observations are rows. At ReLU kinks it uses the zero subgradient. For logits clipped to $[-35, 35]$, it additionally sets the logit derivative to zero outside that interval. Thus it differentiates the actual deployed predictor, rather than an unclipped surrogate.

Because ReLU and clipping are one-Lipschitz and $\sigma' \le 1/4$,

$$\|f_\theta(x) - f_\theta(y)\|_2 \le L_f \|x - y\|_2, \quad L_f = \tfrac{1}{4} L_\Phi \prod_{l=1}^{L+1} \|W_l\|_2. \tag{25}$$

This global bound is conservative; local Jacobians can be much smaller. For coordinate i, the proposal margin derivative is $D\widehat{z}_i = -Df_i/\tau_i$. A useful sensitivity report gives raw units, feature ranges, point derivatives and the spectral bound. It never labels $\widehat{z}_i$ an audited margin. If a separate uniform approximation guarantee $\sup_x |r_i(x) - f_i(x)| \le \epsilon_i$ were established, the triangle inequality would imply $r_i(y) \le f_i(x) + L_f\|y - x\| + \epsilon_i$. A held-out mean log loss does not establish that uniform guarantee.

6.4 A finite-family audit with a stated shift radius

Fix the family, sample sizes and weights before final outcomes are observed. For coordinate i, let $K_i \sim \mathrm{Bin}(n_i, p_i)$ under audit law P_i, and let Q_i be the intended deployment attack law. Assume an externally justified total-variation bound

$$\mathrm{TV}(P_i, Q_i) = \sup_E |P_i(E) - Q_i(E)| \le \rho_i. \tag{26}$$

For weights $w_i > 0$, $\sum_i w_i \le 1$, use the exact one-sided Clopper–Pearson bound

$$U_i = \begin{cases} \mathrm{Beta}^{-1}(1 - \alpha w_i; K_i + 1, n_i - K_i), & K_i < n_i, \\ 1, & K_i = n_i. \end{cases} \qquad U_i^Q = \min\{1, U_i + \rho_i\}. \tag{27}$$

With probability at least $1 - \alpha$, all deployment risks obey $r_i(Q_i) \le U_i^Q$. To prove this, each binomial bound fails with probability at most αw_i; the union bound controls their simultaneous failure. On the complementary event, the definition of total variation applied to the attack-success event gives $r_i(Q_i) \le p_i + \rho_i \le U_i + \rho_i$. No independence across coordinates is needed. Independent Bernoulli trials within each coordinate, a fixed sample size and a fixed target law are needed. Random data-dependent radii require their own simultaneous error budget. A feature displacement and a model Lipschitz bound do not justify a TV radius.

The shifted audited position is $z_i^Q = 1 - U_i^Q/\tau_i$; all must be nonnegative. This is a conditional statistical guarantee given the external shift model, not proof that the radius is true or that unregistered harms are absent. At zero successes, $U_i = 1 - (\alpha w_i)^{1/n_i}$, so passage requires

$$n_i \ge \left\lceil \frac{\log(\alpha w_i)}{\log(1 - \tau_i + \rho_i)} \right\rceil, \qquad 0 \le \rho_i < \tau_i. \tag{28}$$

If $\rho_i \ge \tau_i$, zero observed successes cannot supply a finite passing budget. Repeatedly inspecting these fixed-size bounds and stopping at passage invalidates their stated coverage; use the existing sequential gate instead.

6.5 Audit cost and scaling

For positive costs c_i, set $a_i = c_i/[-\log(1-\tau_i+\rho_i)]$. Ignoring integer ceilings, minimize $C(w) = \sum_i a_i \log(1/(\alpha w_i))$ over positive weights summing to one. The Lagrangian derivative is $-a_i/w_i + \nu = 0$; summing gives

$$w_i^* = a_i / \sum_j a_j. \tag{29}$$

The Hessian is diagonal with entries $a_i/w_i^2 > 0$, proving a unique minimum. The integer plan rounds upward and checks the exact zero-success inequality. It is a planning rule conditional on zero successes, not a promised audit cost under positive risk and not an optimizer of the discontinuous integer objective. Freeze weights from costs and declared tolerances before audit outcomes.

Feature construction costs $O(Nd2^J)$. A dense network forward pass costs $O(N\sum_l d_{l-1}d_l)$; storing every raw-input sensitivity additionally costs $O(Nmd)$ for the output alone. The implementation supports arbitrary d but limits J to eight to avoid accidental dictionary growth. Dimension-free statistical safety does not imply dimension-free computational or predictive accuracy. LLM scenario injections continue through the existing review and measured-development-evidence pipeline; modifying a dictionary or threat law requires a new proposal-model version and new final audits.

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